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Life Cycle Inventories of Packagings & Graphical Papers

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Life Cycle Inventories of Packagings & Graphical Papers

Subprojects "Packagings & Graphical Papers"

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Part I

Introduction

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Final report of the project of a National Life Cycle Inventory Database "Ecoinvent 2000" commissioned by BUWAL/BFE/ASTRA/BLW

Preface to Version v2.0

This report corresponds to the original report (Hischier (2004)) that has been updated with the changes reported in Frischknecht et al. (2006) as well as some additional errors, identified since the publication of the latter report.

In addition, the following major adaptations / changes have been established during the time from 2005 to 2007 within this report here:

- **Plastics** data (part II of this report): the most recent version of the data from PlasticsEurope, the former APME (Association of Plastic Manufacturers in Europe), are used – published in the years 2005 to 2007 (Boustead (2005-07));
- **Corrugated board** data (part III of this report): the most recent version of the data from FEFCO (FEFCO et al. (2006)) replace the former version (FEFCO et al. (2000)) of this unit process-based documentation of the European corrugated board production;
- **Paper sorting** data (part III of this report): a revised version of the paper sorting process, based on personal communications with the paper and board industry, has been established here.

Acknowledgement

The first steps for the ecoinvent project have been taken during the late 1990ties. In the first phase of the project financing and its positioning in existing institutions were negotiated. We thank all the persons involved in this phase who helped to start up this project.

We thank the institutions which supported the development of the ecoinvent database and the investigation of the ecoinvent data. The individual projects for data harmonisation and compilation have been funded by the Swiss Agency for the Environment, Forests and Landscape (BUWAL), Swiss Federal Office for Energy (BFE), Swiss Federal Office for Construction and Logistics (BBL), Swiss Federal Roads Authority (ASTRA), and the Swiss Federal Office for Agriculture (BLW).

Specific acknowledgements for the various datasets established within this project can be found in the respective chapters (part II for plastics, part III for paper and board, part IV for packaging glass).

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1 Introduction

This part of the whole ecoinvent project is not less than a revision and expansion of the famous “BUWAL 250” reports about “Life-Cycle Inventories for Packagings” (Habersatter et al. (1998)). That study itself has already been the revision and expansion of the Environmental Series No.132, “Ecobalance of Packaging Materials, State of 1990” (Habersatter (1991)).

This new version of Life Cycle Inventories of different packaging materials as well as graphical papers is established within the framework of the Swiss ecoinvent project (for a general overview about ecoinvent, see Frischknecht et al. (2007)). Within this framework, the EMPA has taken the responsibility for those inventory data that belong to the packaging and/or graphical paper area. While EMPA in Dübendorf established the wood and metal data, EMPA in St. Gallen took the responsibility for the datasets in the areas of plastics, paper/board and packaging glass.

The whole ecoinvent project was commissioned by several Swiss Federal offices (Swiss Agency for the Environment, Forests and Landscape (BUWAL), Swiss Federal Office for Energy (BFE), Swiss Federal Office for Construction and Logistics (BBL), Swiss Federal Roads Authority (ASTRA), and the Swiss Federal Office for Agriculture (BLW)) as well as the Swiss Centre for Life-Cycle Inventories (ecoinvent centre) itself. The ecoinvent centre under the lead of EMPA is a close co-operation between LCA-institutes in the ETH domain (Swiss federal Institutes of Technology (ETH) Zürich and Lausanne, Paul Scherrer Institute (PSI) Villigen, Swiss Federal Institute for Environmental Science and Technology (EAWAG), and Swiss Federal Laboratories for Materials Testing and Research (EMPA) St. Gallen and Dübendorf) as well as the LCA-department of the Swiss Federal Research Station for Agroecology and Agriculture (FAL) in Zürich.

The famous BUWAL 250 study (Habersatter et al. (1998)) – available not only in German, but also in English and French – has been one of the most used reference documents concerning the life-cycle inventory of packaging materials, in Switzerland but also in the rest of Europe. A part of the European LCA software even contains BUWAL 250 data in their default version. This wide distribution together with the numerous LCA studies referring to these datasets, shows the importance that this work had in the international LCA community.

Once again, like in the project that resulted in BUWAL 250, the main objective of the work within the framework of the ecoinvent project is to update and adapt the data to the state of the art production technologies. As reference year for the data, the year 2000 has been chosen. A second important aspect within this project is the harmonization with all other inventories established within the ecoinvent project. Details about this objective can be found in the methodology report (Frischknecht et al. (2007)).

The data collection was done in close cooperation with companies producing the here further investigated materials and products. Further data sources like other databases, literature or information on the internet was used in cases where no cooperation with respective production sites was possible.

The life cycle inventories were calculated within the ecoinvent database, using a matrix-inversion approach. More information about the calculation as well as the technical background of the ecoinvent database can be found in Frischknecht et al. (2007) and Hedemann (2007).

2 Packaging and the Environment

The following information is mainly taken from Habersatter et al. (1998).

Virtually all products on the market appear in some form of packaging. Packagings encase and protect, define and inform, profile and identify. Storage, transport, self-service and food storage require a protective and supportive wrapper. A packaging which does not sufficiently protect the product is questionable from the ecological viewpoint, since the contents which must be protected virtually always have a higher resource and energy value than that of the packaging. Consequently, the packaging is not an end in itself (rare exceptions to this rule exist), rather it is a necessary means. In addition, each packaging decision is also one of logistics, for it is only the packaging that allows the reliable distribution of goods through space and time in our hectic world. All told, the packaging makes a valuable contribution to the supply of the product and thus occupies a noteworthy position in the value added chain.

However, examination of such a broadly defined performance of packagings is not the topic of this publication "Life Cycle Inventories of Packagings", which does not assess the above-mentioned total (national) economic utilities. Rather, the intention is to illustrate the environmental burden of the manufacture, processing, use and waste treatment of packaging materials raw and processed: the so-called life cycle inventory analysis, established according to the input/output method. This process-related and hence objective tabulation systematically traces the material and energy flows through three aspects of interest for process optimisation. The environmental burden arises from:

- the actual manufacturing processes;
- the energy delivery processes;
- the transport processes.

BUWAL 250 was the first published life cycle inventory analysis of a large number of packaging materials which showed these action-oriented representations. With the complicated processes of BUWAL 250 it was thus possible to see to what extent and why the environmental compartments air and water, as well as the material and energy resources, are burdened. The BUWAL 250 publication made it clear where priority must be given to initiating improvements - aimed at increasing the life cycle efficiency (the central demand of the UNCED of Rio) and introducing lasting economies. Like BUWAL 250, the ecoinvent project and its publications arose through close cooperation between authorities, the scientific community and industry - the reliability and accuracy of both studies are on a correspondingly high level.

Like already BUWAL 250, this project here is a creditable contribution to the objectification of the debate between ecology and business, politics and society in the direction of e.g. an Integrated Product Policy (IPP).

3 Historical overview

The following information is mainly taken from Habersatter et al. (1998).

In the late 1960s various projects using global development models were initiated. The main topics concerned population development and the associated consumption of resources (e.g. Meadows et al. (1972) or SETAC (1991)). The results of these studies showed among other things that fossil energy sources were scarce and would probably last only for a few decades, and that the environmental interventions associated with their consumption could cause climatic changes. These alarming predictions led to a series of detailed energy flow calculations in the field of industrial processes, above all for cost comparisons with alternative energy systems. The investigations concentrated primarily on crude oil processing and steel production as well as on the pulp and paper industry. According to SETAC (1991), at the start of the 1970s in the USA around a dozen such "energy analyses" were performed.

During the oil crisis of 1973-1975, large-scale studies were authorised by the American and British governments covering a wide range of topics concerning industrial systems. As energy flow analyses also require mass balances, these studies also included data on material inputs and wastes. These expanded energy analyses were called "Resource analysis" or "Resource and environmental profile analysis (REPA)". Worthy to be mentioned in this context is the work of Boustead (1972), Franklin & Hunt (1974) or of Sundström (1973). These scientists are still active in this field today and can be regarded as the founders of life cycle assessment.

After the oil crisis, general interest in this type of study waned. Nonetheless the activities were continued, above all in the USA, England, Scandinavia, Germany, the Netherlands and Switzerland.

In the 1980s, sustained by the "green movement" and the associated establishment of environmental organisations, the topic again became a focal point of interest. The packaging industry came under special fire. In particular beverage packagings, which came on the market as disposable containers, were considered wasteful. In the course of an EU Directive on beverage packagings, several studies were performed on energy and material flows. These were used as the basis for corresponding packaging charges.

The 1980s also saw the incorporation of emissions to the atmosphere and water in the energy and material flows in the investigations as a result of an increased environmental consciousness and understanding of the relationships between emissions and destruction of the ozone layer, the greenhouse effect, etc. As a consequence of this expansion the approach, new expressions were coined for this type of study, e.g. product line analysis, eco-balance, life cycle analysis or life cycle assessment.

Motivated by the REPA studies in the USA, the methodology of the "Analysis of environmental assessments" was investigated by EMPA under the leadership of P. Fink. The study "Ecobalances of Packaging Materials" produced by EMPA and published in 1984 by BUS (Bundesamt für Umweltschutz, the former Swiss EPA) as Environmental Series No. 24 (BUS (1984)) aroused great national and international interest and was widely used for the ecological assessment of packagings. In this work an assessment method developed by the Environmental Protection Agency (EPA) in the USA was presented, which allows the numerous parameters of a life cycle assessment to be reduced to a few characteristic values. In 1991 BUWAL published the study after revision and expansion by the ETHZ as BUWAL 132 (Habersatter (1991)). The translations into English and French aroused great international interest.

The life cycle assessment studies conducted in the last 20 years (including their forerunners: REPA etc.) concentrated on the packaging sector. In most of the life cycle analyses of products, waste treatment was dealt with only superficially or even neglected. In the mid-90s, the focus has shifted to investigations of construction materials, inks and varnishes, cleaning agents and detergents, foodstuffs, textiles, transport and energy systems. The integration of so-called downstream processes such as the treatment of waste water and waste in life cycle assessment studies has also been initiated about in the same epoch.

The methods of life cycle assessment (used in what follows as a collective term) have been developed by the respective authorities or in the countries involved virtually completely independently. This has led to numerous different approaches which makes it more difficult to compare the different investigations. For several years, the Society of Environmental Toxicology and Chemistry (SETAC) has been attempting to contribute to the harmonisation of the methods by bringing together experts in the field of life cycle assessment. These efforts are of great importance in general and in particular for the standardisation attempts of the ISO and other bodies.

In addition to the life cycle assessments, which refer primarily to product systems, since the start of the 1990s the activities of the Swiss association for ecologically aware corporate management, ÖBU, have included assessments relating to corporations. The idea of corporate, environmentally related assessments owes its origin to the work of Müller-Wenk (1978). Based on his approach of ecological scarcity, an impact assessment method was developed by Swiss EPA (the method of ecological scarcity, "Umweltbelastungspunkte", UBP) which allows corporations to express the environmental burdens related to the products or operations for which they carry responsibility in terms of a single numerical value. A detailed procedure for the life cycle assessment of corporations based on ecological scarcity has been published by Braunschweig & Müller-Wenk (1993).

Many of the studies conducted to date are not accessible to the general public as they are subject to various confidentiality provisos. As a result, the basic data for many potential application purposes are not available at the moment. In order to remedy this, studies have been initiated by public authorities or industrial associations in Switzerland and also in other Western European countries with the aim of providing a wide circle of users with ecological information on industrial processes. The work of Frischknecht et al. (1994), Boustead (1993), Habersatter (1991) or Habersatter et al. (1998) are examples among such studies.

The next problem that raised up was a sort of compatibility problem when trying to use datasets from various of the above mentioned data sources. Often these different studies have not used exactly the same methodology for their datasets – thus, it is most of the time not possible to compare datasets from two of these studies. With the foundation of the ecoinvent centre, the Swiss authorities together with the LCA research groups from different institution within the ETH domain, a attempt to go ahead one more step is established. Then with ecoinvent, for the first time, such a huge amount of datasets is available in one harmonized database.

4 Content of this report

This report contains, besides the general introduction (part I), information about three of the most important materials in the context of packaging – plastics (part II), paper / board (part III) and glass (part IV). Not included into this report are metals – the datasets for aluminium and steel production can be found in the metals report of ecoinvent (Classen et al. (2007)).

All these life cycle inventories serve as building blocks to generate life cycle assessments for packagings. The data and its report are therefore aimed first and foremost at people concerned with packagings. They include packaging manufacturers, major distributors, authorities or consultants and engineering firms. The data in this report here concentrates – like in ecoinvent in general – on Swiss and Western European conditions. While applications in relatively large economic areas are possible, each individual case must be checked as not all production locations are comparable with regard to technology, energy provision or means and types of transport.

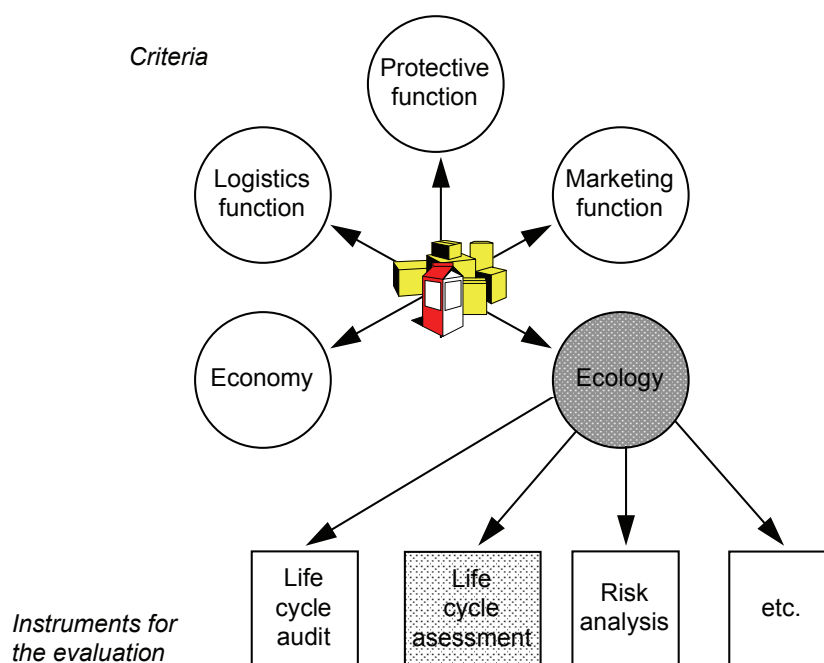


Fig. 4.1 Decision criteria for packagings and instruments for their ecological evaluation (Fig. 5.1 from Habersatter et al. (1998))

Various aspects are used in the evaluation of a packaging solution. Packaging has to fulfil protective functions which include not only protection of the product (e.g. perishable goods), but also protection of the environment (e.g. against hazardous goods). It must meet demands regarding storage and transport and serve as an information source (e.g. declaration). In addition to these three functions, the two main aspects of economy and ecology are decisive for the evaluation. Decision criteria in this context are shown in Fig. 4.1.

Environmental influences can be evaluated in widely different ways, e.g. with life cycle audits, life cycle assessments, risk analyses and many others. The different instruments differ in their application and provide different information (see also Beck & Bosshart (1995)). The life cycle assessment determines and evaluates environmental influences over the entire life cycle of a product or activity. It is based on input-output analyses, i.e. material and energy assessments over the life cycle defined as the system.

Life cycle inventories can be used as a basis for environmentally relevant decisions in the packaging field whether this involves selecting a packaging system, developing new packagings or optimising existing solutions.

The whole ecoinvent project has a modular structure. While each step shows the direct relations with the technosphere (energy, auxiliaries etc. used – services for disposal of waste) and the biosphere (direct emissions to air and water), the calculated cumulative data cover then the entire process chain from raw material acquisition, including all transport, up to the factory gate: in other words "**from cradle to gate**". The "gate" is characterised by the last process stage considered in the inventory and its location. The following list gives an overview of the materials, usable for the production of packaging systems, described within this report here:

	Materials	Gate	
		Process step	Location(s)
	Plastics • Polyolefins (HDPE, LDPE, LLDPE, PP) • Polystyrene (GPPS, HIPS, EPS) • Chlorinated Polymers (PVC, PVDC) • Polyethylene Terephthalate (PET) • Polymethyl Methacrylate (PMMA) • Polyurethanes • ABS / SAN • Polycarbonate • Nylon (type 6 & 66)	Polymerization plant Polymerization plant Polymerization plant Polymerization plant Polymerization plant Polymerization plant Polymerization plant Polymerization plant Polymerization plant	Europe Europe Europe Europe Europe Europe Europe Europe Europe
	Graphical papers • Newsprint • Wood-containing paper (SC, LWC) • Wood-free paper (coated, uncoated) • Recycling (with / without deinking)	Paper mill Paper mill Paper mill Paper mill	CH, Europe Europe Europe Europe
	Packaging papers • Bleached kraft paper • Unbleached kraft paper	Paper mill Paper mill	Europe Europe
	Corrugated Board • Corrugated base paper (Semichem. Fluting, Kraftliner, Wellenstoff, Testliner) • Corrugated board (single/double wall; fresh/mixed/recycling fibre)	Paper mill Paper mill	Europe CH, Europe
	Cartonboard • Folding box board (FBB) • Whitelined chipboard (WLC) • Solid bleached board (SBB) • Solid unbleached board (SUB) • Liquid packaging board (LPB)	Paper mill Paper mill Paper mill Paper mill Paper mill	Europe Europe Europe Europe Europe
	Packaging glass • white packaging glass • green packaging glass • brown packaging glass	Glass bottle production Glass bottle production Glass bottle production	CH, DE, Europe CH, DE, Europe CH, DE, Europe

Specific processes for the production of packaging out of these raw materials are included for the plastics as well as parts of the paper / board sector.

In cases where the packaging raw material is produced out of an intermediate product, these intermediate products are also included into the report here. Examples therefore are the monomers of the various plastics mentioned above (e.g. ethylene, styrene, vinyl chloride, etc.) or the different types of market pulp (e.g. ECF sulphate pulp, TMP pulp, etc.).

Specific auxiliaries that are mainly used for one of the mentioned packaging materials are also reported in the mentioned part of the report and are not part of the chemicals report (Althaus et al. (2007)), while general auxiliaries can be found in the mentioned chemical report (e.g. kaolin as a filler is reported within part III of this report here – sodium hydroxide is reported in Althaus et al. (2007)). For the additional datasets like energy carriers, transport means and waste disposal, data from the respective ecoinvent sub-projects are used. Thus, these data are reported not within this report here but in the respective other reports out of the ecoinvent report series (e.g. waste disposal is described in Doka (2007)).

Nevertheless, the present life cycle inventories do not represent the entire life cycle of a packaging. To obtain such a life cycle, the user must link the modules available in this report here with one another, and determine the missing modules for a particular case. It is only in this manner that the linked modules constitute a precise life cycle inventory for a specific packaging.

5 Position / Application of this report

The following information is mainly taken from Habersatter et al. (1998). The life cycle assessment permits various applications, such as a weak point analysis or a comparison of system variations. Application possibilities in the field of design or in the development of products are also conceivable.

- **Weak point analysis:** This allows an ecological assessment of various manufacturing processes in one's own plant or of upstream or downstream processes. The main question is: where do the major environmental burdens appear? Among other things, the influence of transport routes on the entire environmental burden of the corresponding production can be investigated.
- **System variations:** A comparison of different variations allows different packaging materials or packaging types with the same functional unit to be compared. Various questions, such as the influence of different recycling rates of a product on the environmental burden, can thus be answered.

The reliability of a life cycle assessment depends essentially on the care with which it is performed. The result of a life cycle assessment must be transparent and traceable. This requires observance of the following basic rules:

- The system and the system boundaries must be clearly defined. This should be made clear by use of a flow chart.
- All assumptions and simplifications must be clearly apparent.
- The functional unit must be explicitly defined.
- The existing gaps must be explicitly shown.
- The methodological decisions (allocation rules, valuation models etc.) should be disclosed.
- The literature or data basis must be specified.

To perform life cycle assessments, recognised guidelines (such as those of the ISO 14'040 Series) should be followed.

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